



Trimble India AI Summit

**Guardrails at the
Speed of AI**

*A Privacy-Safe, Carbon-Aware Middleware
for Responsible Agent Orchestration*



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01

The Problem

No One Is Measuring

Problem Statement

Guardrails for AI Agents in Trimble

The Governance Gap

“Securing the AI Lifecycle Against Sensitive Data Exposure, Environmental Impact, and Ethical Drift”





Existing Data Integrity Problem

- Trimble is undergoing **large-scale AI adoption**, leveraging tools such as Gemini, Cursor, GitHub Copilot, and Agentic AI platform to drive enterprise-wide transformation.
- As AI usage intensifies across the organization, critical oversights around **sensitive data exposure** to models are becoming an increasingly significant risk.
- Beyond data privacy, the **carbon footprint of AI** — driven by inefficient or uninformed prompting practices — remains an underappreciated consequence of scaling without awareness.

*As **Trimble** accelerates its AI journey, the path forward is clear: **scale intelligently and responsibly** — balancing innovation with governance, sustainability,*

The Problem



Data Leakage by Design

AI agents ingest employee names, IDs, credentials & API keys. Without interception, sensitive data flows unchecked into model contexts, logs, and outputs — compliance exposure at every call.



Invisible Carbon Debt

Every AI interaction carries a compute cost. At enterprise scale, continuous inference pipelines generate unmeasured carbon. Without budget-aware routing, energy scales linearly with zero accountability.



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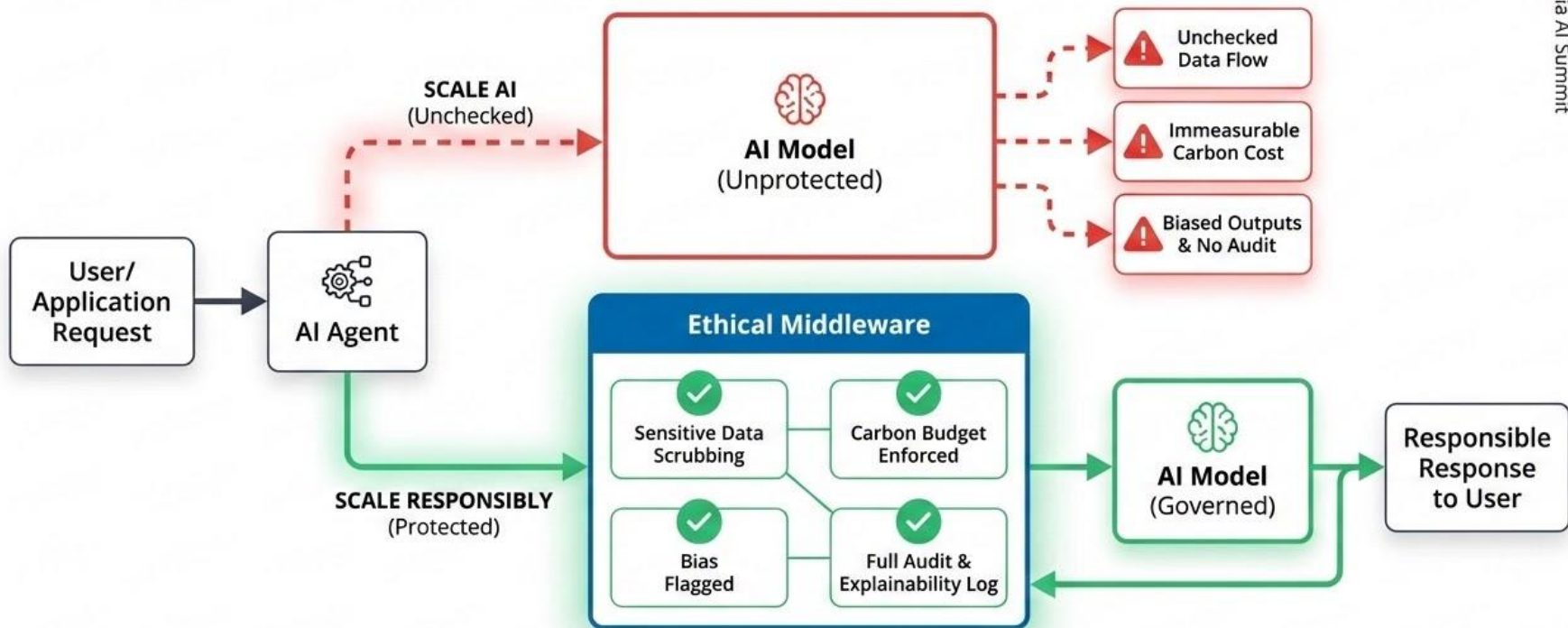
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The Solution

Ethical Middleware Layer

The Solution

Ethical Middleware as a Service



WITHOUT MIDDLEWARE

"Assign crane operator John Smith (EMP-4521, john.smith@trimble.com) to Site B."

- ⚠ Sensitive data persists in model context
- ⚠ Employee ID & email logged in analytics
- ⚠ High-compute model used for simple routing
- ⚠ Deployment action proposed — no record of energy used
- ⚠ Zero audit trail of what data was exposed

WITH MIDDLEWARE

*"Assign crane operator EMP-4521, john.smith@trimble.com to Site B." →
Warning that sensitive data is exposed*

- ✓ Sensitive data scrubbed — never reaches model
- ✓ Carbon classified: query = 0.001 kg CO₂, allowed
- ✓ Response scanned — no leakage detected
- ✓ Deployment action checked against carbon budget
- ✓ Blocked if over budget — full log maintained



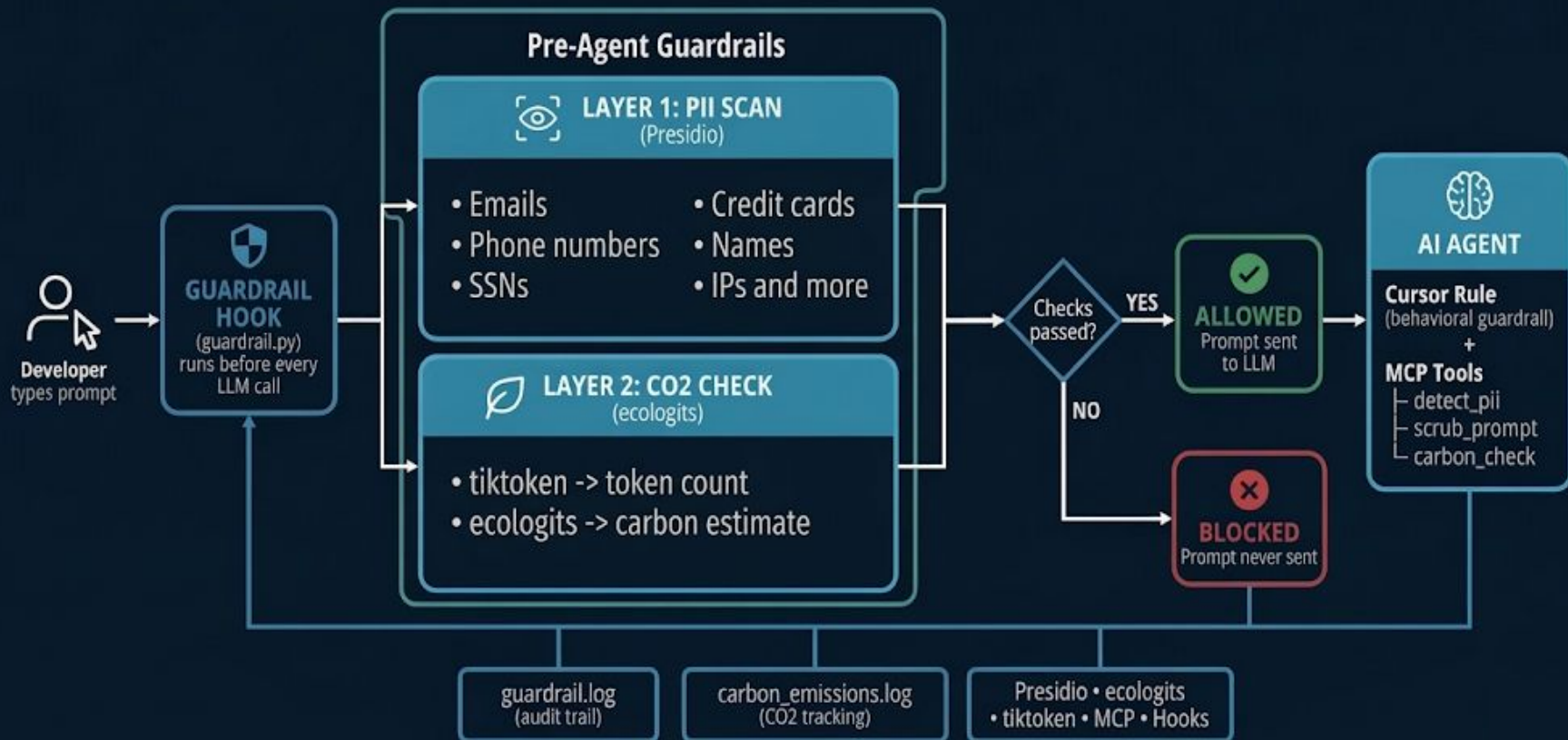
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Architecture

One Middleware, Every Agent



Architecture



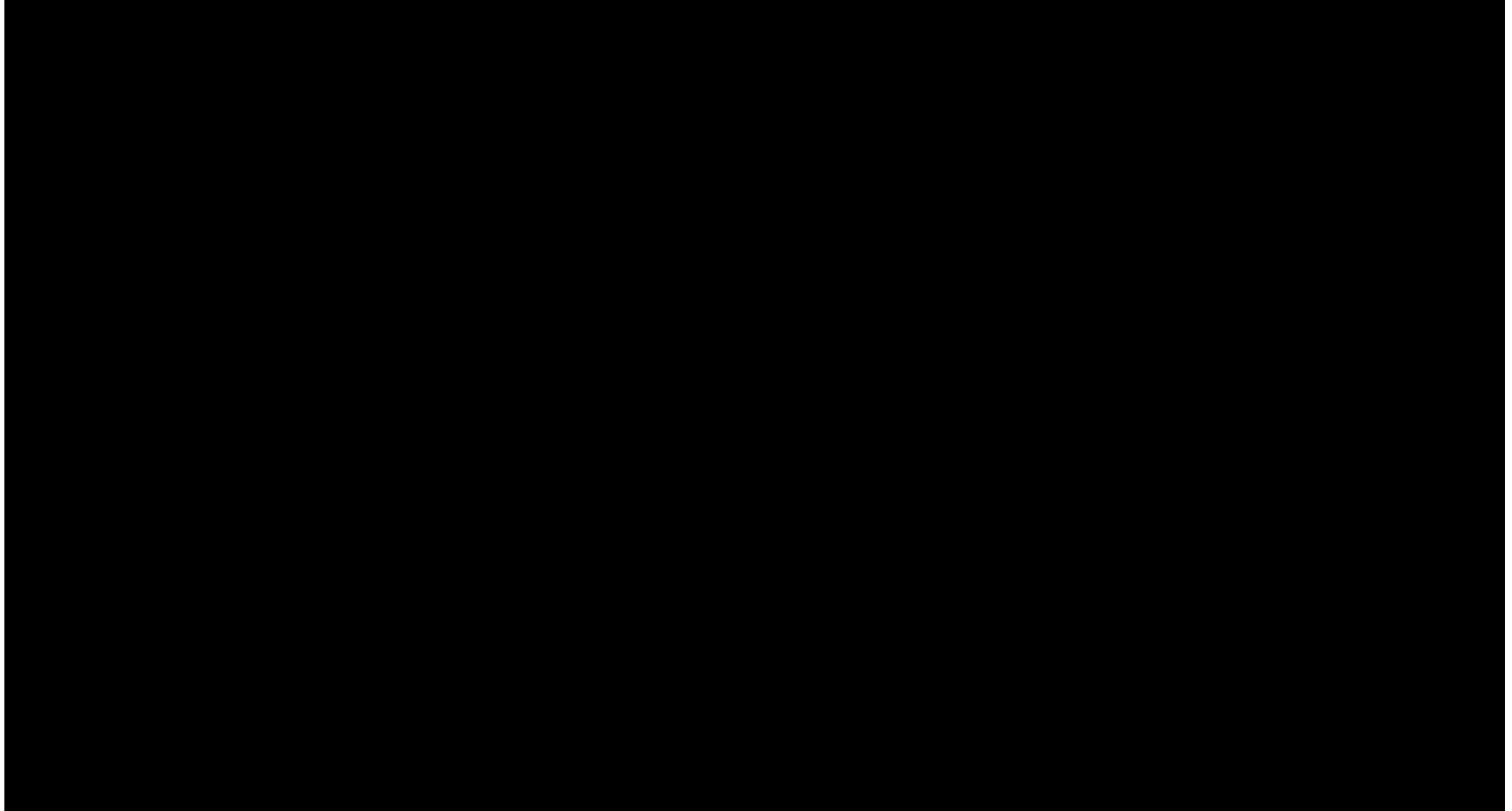
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Demo



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Demo



Why this Governance Layer?



Risk Reduction

Eliminates sensitive data exposure across all AI-powered workflows — from field operations to geospatial analytics.



Scalable Governance

One middleware layer governs all agents, replacing fragmented per-team controls with a single audit-ready standard.



Industry Trust

Demonstrates Responsible AI in action for customers in construction, agriculture, and transportation.

05

Impacts



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Meaningful Impacts

~60%

Reduction of AI **data breaches** and
prompt injection

~50%

Reduction of organization's AI
carbon footprint

06

Future Scope



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Future Scope



Extend to Output Guardrails

Scan AI agent's output for security vulnerabilities, potential data loss before executing shell commands.



Custom Trimble Governance

Implement Trimble specific Governance policies for handling AI interactions



Handle Data Masking for smooth AI experience

Instead of blocking prompts, use an additional abstract layer that utilizes cache to mask and de-mask sensitive data without disrupting user's interaction with the agent.



Optimal Prompt Recommendations

Provide recommendations for optimal prompts that can reduce carbon emissions



Scale to other AI systems

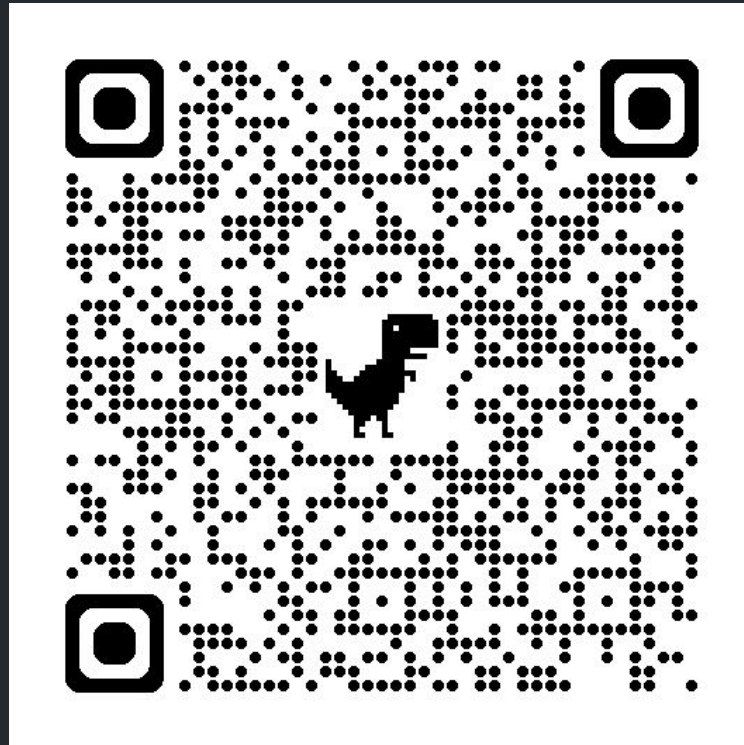
Extend this implementation beyond Cursor and incorporate industry-specific guardrails in AI systems across Trimble.



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Thank You!!

Feedback





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